Algorithm #5 The Insertion Sort.

Procedure insertion sort ($a_1, a_2, \ldots, a_n$: real numbers with $n \geq 2$)

for $j := 2$ to n — start $j = 2$ until n

  $i := 1$

  while $a_j > a_i$

    $i := i + 1$ } Step #1: Find i

  $m := a_j$ ← Step #2 Save $m = a_j$

  for $k : 0$ to $j - i - 1$

    $a_{j-k} := a_{j-k-1}$ } Step #3: Shift a_{j-k} moves to a_{j-k-1} if $j-1-1 \geq 0$

  $a_i := m$ } Step #4: Insert $a_i = m$

{ $a_1, \ldots, a_n$ is in increasing order }

Example: We will sort $[3, 19, 8, 9, 12, 5, 4, 1]$, $n = 8$

for $j = 2$ to n

1). Find position i where a_j belongs

2). Save $\boxed{m := a_j}$

3). Shift elements right using:

for $k : 0$ to $j - i - 1$

$\boxed{a_{j-k} := a_{j-k-1}}$

4). Insert: $\boxed{a_i := m}$

Note:

while $a_j > a_i$ — If TRUE → move forward; If FALSE → stop

  $i := i + 1$

The loop continues while the condition is TRUE

The loop stops when the condition becomes FALSE.

Iteration $j=2$

Array: $[3, 19, 8, 9, 12, 5, 4, 1]$ $(a_2 = 19)$

1) Step 1: Find i

Start $i = 1$

$19 > 3$ ✓

$19 > 19$ ✗ Stop $\rightarrow i = 2$

2) Step 2: Save

$\boxed{m = 19}$

3) Step 3: k-loop $\quad j - i - 1 = 2 - 2 - 1 = -1$

No shifting. Array unchanged. $[3, 19, 8, 9, 12, 5, 4, 1]$

Iteration $j=3$

Array: $[3, 19, 8, 9, 12, 5, 4, 1]$ $(a_3 = 8)$

1). Step 1: Find i

Start $i = 1$

$8 > 3$ ✓ $\rightarrow i = 2$

$8 > 19$ ✗ stop. So $i = 2$

2) Step 2: Save

$\boxed{m = 8}$

3). Step 3: k-loop $\quad j - i - 1 = 3 - 2 - 1 = 0$

k runs from 0 to 0 (1 iteration)

When $k = 0$

$$a_{3-0} = a_{3-0-1}$$

$$a_3 = a_2$$

Array becomes: $[3, 19, 19, 9, 12, 5, 4, 1]$

$a_2 = a_3$

4) Step 4: Insert

$\boxed{a_2 = 8}$

Array: $[3, 8, 19, 9, 12, 5, 4, 1]$

Iteration $j=4$ ($a_4=9$) Array: [3, 8, 19, 9, 12, 5, 4, 1]

1) Step 1: Find i

$9>3$ ✓

$9>8$ ✓

$9>19$ ✗ Stop $\rightarrow i=3$

2) Step 2: Save

$\boxed{m=9}$ $(m=a_j=a_4=9)$

3) Step 3: k-loop $k=\overset{j}{4}-\overset{i}{3}-1=0$ (1 iteration)

$a_{j-k}=a_{j-k-1}$

$a_4=a_3$

Array [3, 8, 19, 19, 12, 5, 4, 1]

$a_4=a_3$

4) Step 4: Insert

$\boxed{a_3=9}$

Array [3, 8, 9, 19, 12, 5, 4, 1]

Iteration $j=5$ ($a_5=12$) Array [3, 8, 9, 19, 12, 5, 4, 1]

1). Step 1: Find i

$12>3$ ✓ $12>8$ ✓ $12>9$ ✓ $12>19$ ✗ $\rightarrow i=4$

2) Step 2: Save $\boxed{m=12}$ $(m=a_j=a_5=12)$

3) Step 3: k-loop $k=j-i-1=5-4-1=0$ (1 iteration)

$a_{j-k}=a_{j-k-1} \rightarrow a_5=a_4$

4) Step 4: Insert $\boxed{a_4=12}$

Array [3, 8, 9, 12, 19, 5, 4, 1]

Iteration $j=6$ $(a_6 = 5)$ Array $[3, 8, 9, 12, 19, 5, 4, 1]$

1) Step 1: Find i

$5>3$ $5>8$ × Stop $\rightarrow i=2$

2). Step 2: Save $\boxed{m=5}$ $(m = a_j = 5)$

3). Step 3: k-loop $k = j-i-1 = 6-2-1 = 3$

$k=0$ to 3

$k=0$ $a_6 = a_5$ → move 19

$k=1$ $a_5 = a_4$ → move 12

$k=2$ $a_4 = a_3$ → move 9

$k=3$ $a_3 = a_2$ → move 8

Array becomes: $[3, 8, 8, 9, 12, 19, 4, 1]$

a_2 a_2 a_3 a_4 $a_6 = a_5$

locations of a_2 & a_3

(after all shifts)

4) Step 4: Insert $\boxed{a_i := m}$ or $\boxed{a_2 = 5}$ Array $[3, 5, 8, 9, 12, 19, 4, 1]$

Iteration $j=7$ $(a_7 = 4)$

1). Find i $4>3$ $4>5$ × Stop → $i=2$ 2) Save $\boxed{m=4}$

3). Shift $7-2-1=4$ shift 19, 12, 9, 8, 5 one step right.

4) Insertion 4 at position 2 $\boxed{a_2=4}$ $[3, 4, 5, 8, 9, 12, 19, 1]$

Iteration $j=8$ $(a_8 = 1)$

1). Find i $1>3$ × Stop → $i=1$ 2). Save $\boxed{m=1}$

3). Shift $8-1-1=6$ shift 19, 12, 9, 8, 5, 4, 3 right (one step)

4). Insert 1 at position 1 $\boxed{a_1=1}$ $[1, 3, 4, 5, 8, 9, 12, 19]$